Accepted Manuscript

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PII: S0165-0270(18)30045-1
DOI: https://doi.org/10.1016/j.jneumeth.2018.02.015
Reference: NSM 7953

To appear in: *Journal of Neuroscience Methods*

Received date: 9-1-2018
Revised date: 13-2-2018
Accepted date: 20-2-2018

Please cite this article as: López-Blanco Roberto, Velasco Miguel A, Méndez-Guerrero Antonio, Romero Juan Pablo, del Castillo María Dolores, Serrano J Ignacio, Benito-León Julián, Bermejo-Pareja Félix, Rocon Eduardo.Essential Tremor Quantification Based on the Combined Use of a Smartphone and a Smartwatch: the NetMD study.*Journal of Neuroscience Methods* https://doi.org/10.1016/j.jneumeth.2018.02.015

Essential Tremor Quantification Based on the Combined Use of a Smartphone and a Smartwatch: the NetMD study

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Running Title: Essential tremor quantification using a smartphone and a smartwatch.

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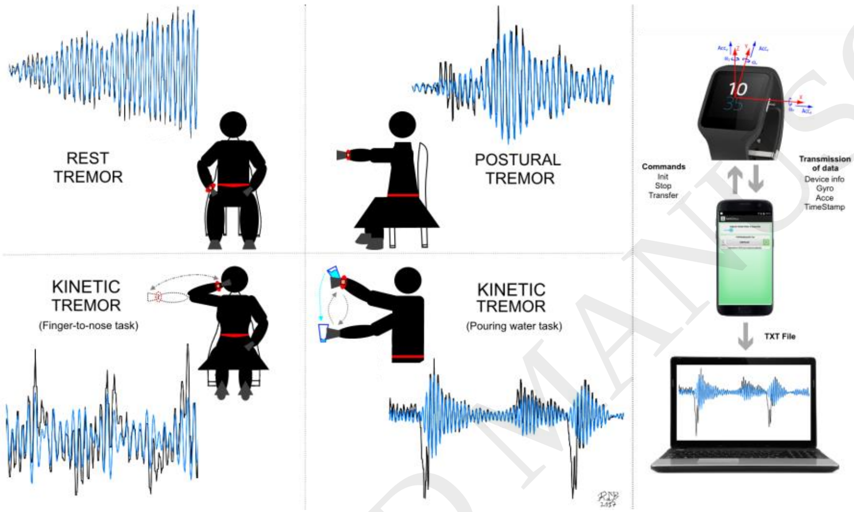
Graphical Abstract



Highlights

- Smartwatches can be adapted to register tremor in essential tremor patients.
- Gyroscopes integrated in smartwatches are useful for tremor quantification.
- Good correlation was shown between tremor rating scale and smartwatch gyroscope data analysis.
- Test-retest reliability in tremor intensity measures was trustworthy.
- This wearable obtained a good acceptation by essential tremor patients.

ABSTRACT.

Background: The use of wearable technology is an emerging field of research in movement disorders. This paper introduces a clinical study to evaluate the feasibility, clinical correlation and reliability of using a system based in smartwatches to quantify tremor in essential tremor (ET) patients and check its acceptance as clinical monitoring tool. **New Method:** The system is based on a commercial smartwatch and an Android smartphone. An investigational Android application controls the process of recording raw data from the smartwatch three-dimensional gyroscopes. Thirty-four ET patients were consecutively enrolled in the experiments and assessed along one year. Arm tremor was videofilmed and scored using the Fahn-Tolosa-Marin Tremor Rating Scale (FTM-TRS). Tremor intensity was quantified with the root mean square of angular velocity measured in the patients' wrists. **Results:** Eighty-two assessments with smartwatches were performed. Spearman's correlation coefficients (ρ) between clinical tremor (FTM-TRS) scores and smartwatch measures for tremor intensity were 0.590 at rest; ρ=0.738 in steady posture; ρ= 0.189 in finger-to-nose maneuvers; and ρ= 0.652 in pouring water task. Smartwatch reliability was checked by intraclass realiability coefficients: 0.85, 0.95, 0.91, 0.95 respectively. Most of patients showed good acceptance of the system.**Comparison with Existing Method(s):** This commodity hardware contributes to quantify tremor objectively in a consulting-room by customized Android smart devices as clinical monitoring tool. **Conclusions:** The NetMD system for tremor analysis is feasible, well-correlated with clinical scores, reliable and well-accepted by patients to tremor follow-up. Therefore, it could be an option to objectively quantify tremor in ET patients during their regular follow-up.

Abbreviations: ET: Essential Tremor, SW3: Smartwatch3, RT: Rest tremor, PT: Postural tremor, KTn: Kinetic Tremor in finger-to-nose maneuver, KTg: Kinetic tremor in pouring water from a glass, FTM-TRS: Fahn-Tolosa-Marin Tremor Rating Scale; ICC: Intraclass Correlation Coefficient, MDC: Minimum detectable change, CR: Coefficient of repeatability, SEM: Standard error of measurement.

Key Words: Essential tremor; tremor quantification; smartphone; smartwatch; tremor transductor;

INTRODUCTION

Essential tremor (ET) is considered one of the most prevalent movement disorder in adults, affecting ~ 5% of people over age 65 (Benito-Leon and Louis, 2006). Tremor occurs related to posture or movement of the limbs (Deuschl et al., 1998), but mild rest tremor can be observed in patients with long-standing severe tremor (Benito-Leon and Louis, 2006; Cohen et al., 2003).

Validated tremor rating scales are good clinical tools for the evaluation of ET (Elble et al., 2013). However, these clinical assessments are subject to biases from the inter-rater and the intra-rater variability of the examination (Elble et al., 2013). The search for a system that allows us to register tremor by an easy, objective and continuous way in ET patients is challenging for clinicians (Haubenberger et al., 2016). In addition to utility, reliability, validity, and accessibility, wearability and ecology are required features of any new device intended to monitor tremor in controlled and free-living conditions (Del Din et al., 2016).

Different transducers have been used for the quantification of tremor (Elble and McNames, 2016; Haubenberger et al., 2016), such as electromyography (Carboncini et al., 2001), drawing of samples and handwriting by digitizing tablets (Elble et al., 1990), actigraphy with accelerometers (van Someren et al., 1998) or gyroscopes (Gallego et al., 2010), and even by means of laser transduction of velocity and electromagnetic tracking (O'Suilleabhain and Dewey, 2001) or optical devices (Chen et al., 2016). However, most of these methods or systems are time-consuming or impractical to be used in daily life.

The miniaturization of inertial sensors as a part of microelectromechanical systems technology has permitted the design of new wearable wireless inertial measurement units which contain triaxial accelerometers, gyroscopes, and magnetometers. Wearable sensing technology and improvements in the design of machine learning algorithms have recently shown the potential to recognize movement disorders (de Lima et al., 2006; Heldman et al., 2011). Furthermore, they have been used to quantify tremor during standard clinical exams of ET (Mostile et al., 2010) or motor symptoms in Parkinson's disease patients (Giuffrida et al., 2009). Industrial designs of health wearables (Mostile et al., 2010; Tzallas et al., 2014) or even non-health related commercial devices, such as smartphones, smartwatches, or bracelets with customized applications (Kubben et al., 2016; Silva de Lima et al., 2016; Wile et al., 2014) could permit an ecologic measure of tremor. Specifically, non-health commercial devices are more versatile, ubiquitous and easily accessible to consumers at a relative low cost (Harries et al., 2016). However, only a few studies comprising a reduced number of patients have used smart devices to quantify tremor in ET (Kubben et al., 2016; Senova et al., 2015; Zheng et al., 2017).

The aim of this clinical study is to check the feasibility, the clinical correlation and the reliability of tremor quantification using commodity hardware (such as smartwatches) (Figure 1) to develop a tool to monitor ET patients in a consulting-room. This paper also addresses the acceptance of the whole system as a tool for everyday use.

MATERIALS AND METHODS

All procedures were approved by the Ethical Standards Committee on human experimentation at the University Hospital ‘‘12 de Octubre’’ (Madrid). Written (signed) informed consent was obtained from all participants.

Participants

Thirty-four ET patients were consecutively recruited from the outpatient neurology clinics of the University Hospital “12 de Octubre” in Madrid (Spain) after obtaining an appropriate informed consent. None of patients had history of stroke, epilepsy, head injury, pacemaker or brain stimulator. Patients were followed-up on average one year. A total of six patients abandoned the study after recruitment due to familiar troubles or unrelated medical problems.

Apparatus

The NetMD system consists of a Smartwatch3 (SW3) Sony© worn in the wrist of the more tremorous limb, and an Android Smartphone ASUS© inside of a belt-pouch (Kalenji®) placed in the waist. An investigational smartwatch and smartphone NetMD Android Wear OS application designed by NetMD's consortium was used to record raw data obtained from gyroscopes of up to five smartwatches synchronously at sampling frequency of 50Hz, see Figure 1A. In these experiments, a single smartwatch and smartphone were paired through Bluetooth. The smartphone stored a timestamp and the tridimensional (3D) data from the accelerometers and gyroscopes in a text file (txt). Figure 1B depicts the communication process between the smartphone and the smartwatch during the acquisition of the patient's movements.

Procedure

Prior to the experiments, a neurologist with expertise in movement disorders (R.L.B.) examined the patients and used the Fahn-Tolosa-Marin (FTM) tremor rating scale (TRS) to assign a total tremor score (range = 0 – 144) (Fahn et al., 1993). Diagnoses of ET were assigned using the Consensus Statement on Tremor by the Movement Disorder Society (Deuschl et al., 1998). A complete clinical history, a FTM-TRS score, and a Montreal Cognitive Assessment (Moca) score were obtained at recruitment.

All the recruited ET patients underwent an abbreviated videotaped neurological examination focused on tasks eliciting tremor while wearing the NetMD system at recruitment, and 33 patients with adherence to follow-up were re-assessed at second visit (from 3 to 6 months after recruitment), with the same tasks (18 patients wore one smartwatch in one wrist, and 15 patients wore a total of two smartwatches (one in each wrist). The neurologist (R.L.B) examined the videos of each ET patient and scored the items related to rest, postural, and finger-to-nose kinetic tremor, as well as to the task of pouring water from a glass kinetic tremor, according to the FTM-TRS part A and B (Fahn et al., 1993). The examination tasks and score patterns are shown in Supplementary File Tables 1,2. The videos were also rated by a second blinded trained neurologist (A.M.G) according to the same criteria. The average of the scores was calculated and used as the gold standard for the assessment of tremor. This increased the reliability of the clinical score of each item.

The ET patients continued taking medication for their disease during the tests. In addition, all the patients were asked to answer a questionnaire, which was adapted from Giuffrida et al (Giuffrida et al., 2009)), about their acceptance of the system.

Finally, to check reliability of the system, twelve months after recruitment 28 patients were reassessed with the same tasks twice, under similar conditions keeping one smartwatch in one wrist. Test and retest were performed consecutively in the same examination protocol.

Tremor quantification

The txt files were processed on a 2.83 GHz Inter Core 2 Quad Q9500 machine running Windows 7 Professional 32-bit. Segments of the 3D gyroscope signals related to the four clinical tasks were post-processed off-line using Matlab software (version 7.11.0 (R2010b); MathWorks, Natick, MA). Hand tremor of ET patients is generally characterized by rhythmic movements at a frequency between 4 and 12 Hz (Benito-Leon and Louis, 2006). To detect the presence of tremor, the angular velocity of the three axes were analyzed separately. The gyroscope data were band-pass filtered using a 10th order Butterworth high-pass filter (f_1>4 Hz) followed by a 10th order Butterworth low-pass filter (f_2<8 Hz). The former subtracted the volitional movements (<3 Hz); the latter filter, other high frequency tremors, such as orthostatic tremor or enhanced physiological tremor (>8 Hz) according to previous studies (Gallego et al., 2013; Zheng et al., 2017). The intensity of tremor was calculated from the root mean square (RMS) of a 1-second moving window of the gyroscope signals. This window size was chosen to optimize the relation between accuracy and time resolution. The intensity was quantified with the parameter: Mean intensity, *MI* (º/s): the average value of the RMS in the three axes.

The dominant frequency of the tremor, *FT* (Hz), was estimated using the SPECTROGRAM function in Matlab, with a window size of 1 s and a 90 % of overlap.

Statistical analyses

The statistical analysis was performed using the free software Version 1.0.136 – © 2009-2016 RStudio, Inc. A total of eighty-two tremor registries were processed although three of them were removed from the analysis due to corrupted video files. Weighted Cohen's kappa coefficients were obtained between tremor raters for each task. The distributions of intensity of tremor measures and the mean of clinical tremor scores were analyzed using Shapiro-Wilk test of normality. Correlations between measures and scores were obtained for each task using Spearman's rank correlation coefficient.

Smartwatch reliability was assessed by Bland-Altman plots, analyzing additional fifty-six tremor registries from a total of twenty-eight patients who were tested repeatedly twice. Standard error of measurement (SEM), intra-class correlation coefficients (ICC) and coefficient of repeatability (CR), mean differences (bias) for each tasks were analyzed. Oultiers were removed in a first step for calculation repeatability parameters when the value was > 3 standard deviation from mean difference (Parrinello et al., 2016) and clinical change between ratings in FTM-TRS ≥1.

RESULTS

A total of thirty-four ET patients were consecutively recruited and asked to be followed up along one year, obtaining a total of eighty-two smartwatch registries for each task. Subsequently, twenty-eight patients completed a repeated assessment twice under similar conditions, obtaining fifty-six additional tremor registries to check repeatability.

The mean age of the ET patients was 64 years (range 18 to 81). Eighteen patients (52%) were taking medication for their disease (propranolol and/or primidone). The main clinical features of the ET patients are shown in Table 1. A summary of the smartwatch analysis of dominant frequency of the tremor can be found in Supplementary File Table 3.

The distribution of intensity measures are illustrated in Figure 2. Spearman's correlation coefficients (ρ) between clinical tremor (FTM-TRS) scores and smartwatch measures for tremor intensity at first assessment were 0.590 ($p<0.001$) at rest; $\rho=0.738$ ($p<0.001$) in steady posture; $\rho= 0.189$ ($p=0.094$) in finger-to-nose maneuvers; and $\rho= 0.652$ ($p<0.001$) in pouring water task. Smartwatch intraclass realiability coefficients: 0.85, 0.95, 0.91, 0.95 respectively. Also, remaining test-retest reliability parameters are illustrated by Bland-Altmann plot in Figure 3. Standard error of measurement (SEM), coefficient of repeatability (CR), and mean differences (bias) are shown in Table 2.

The questionnaire results showed global acceptance and good wearability of the device in most of the patients with a satisfaction level of 8.7 over 10 (see Table 3).

DISCUSSION

The main objectives of this study were to introduce and to evaluate the feasibility and reliability of using a system based on smartwatches to register tremor in ET patients and check its clinical correlation as a monitoring tool of ET patients in a consulting-room along time. Additionally, we were interested in checking its grade of acceptance.

Rest, postural and kinetic tremor

A logarithmic relationship between the data registered by the transducers data and a standard five-point scale was estimated (Elble et al., 2006; Giuffrida et al., 2009). The NetMD system showed moderate to good correlation with most of the clinical tremor (FTM-TRS) scores in a consulting room. We found a strong correlation in postural tremor and pouring water kinetic tremor scores and moderate in rest tremor.

The kinetic tremor estimated from the data recorder by the gyroscopes in the finger-to-nose maneuver was not correlated with the clinical measures. A possible explanation could be the great variability found in the strategies which ET patients chose to complete the task. They performed the movements at different velocities and many of them had problems keeping the required posture, i.e. a 90º extension of the shoulder. Similar limitations in this item were reported (Mostile et al., 2010) for this item as “finger-to-nose movements can still produce

3.5 to 11 Hz spectral frequency components", which overlap with the tremor signal.

One of our contributions in this paper was to show that this limitation could be overcome with a more standardized task such pouring water from a glass. In fact, we estimated a moderate to good correlation with clinical ratings of kinetic tremor in the task that involved pouring liquids. The agreement between tremor raters was moderate to substantial according to Landis and Koch classification (Landis and Koch, 1977) and the observed variability in clinical scores may be explained in the context of previously described inter-rater variability (Stacy et al., 2007).

Smart devices and clinical practice

This study supports the feasibility of adapting Android wearables as transducers of tremor in a specialized clinical setting to quantify the intensity of tremor in an objective way. Automating this procedure requires the implementation of signal processing algorithms by engineers and the meticulous analysis by neurologists for the configuration and validation of those algorithms. However, after validation this process would be automatic. Also, the system has shown a good reliability in repeated measures, positioning as a good monitoring tool.

Propietary systems such as the KinesiaTM (Giuffrida et al., 2009) or the Wiimote® (Synnott et al., 2011) have been proposed to assess tremor and overall motor function in PD patients. Other platforms, which assess symptoms other than tremor in PD patients (e.g., bradykinesia, rigidity) are also available (Tzallas et al., 2014). ET has received considerably less attention, although some works on tremor quantification and classification during the performance of activities daily living using simple spectral analysis are available (Heldman et al., 2011). To the authors' knowledge, the studies addressing the use of a combination of commodity hardware in the tremor quantification in ET patients are scarce (Zheng et al., 2017) and this is the first follow-up study with smartwatches in a consulting room.

Experimental concerns

The study was not without limitations. The location of NetMD's smartwatch (in the wrist) may impede the detection of distal finger tremor which would be easily detected by visual inspection. This could reduce the accuracy of the system (Elble and McNames, 2016). A more accurate detection and measurement of tremor could be achieved by placing multiple three-dimensional motion sensors (Lambrecht et al., 2014). However, this would lead to a loss of wearability apart from requiring a more complex signal processing. Despite this, the NetMD is an usable system which showed good acceptability during a common tremor exam in a consulting room. The experiments followed a similar protocol than the one used for the validation of other developed devices (Giuffrida et al., 2009; Mostile et al., 2010). The NetMD system can be a valid option to objectively measure the occurrence of tremor and its severity. One of the most appealing features which this concept of analysis provides the clinicians with the freedom and versatility of the Android smart devices. They can be programmed to register movement parameters without trade obstacles. Other advantage is the fact that no specific device is required.

On the other hand, the presence of rest tremor in clinical assessment was higher than the percentage described previously (Cohen et al., 2003) but these figures vary on population-based setting(Louis et al., 2015). It was observed in about a third of patients but always with predominant postural and kinetic tremors. Unexpectedly, the analysis of the smartwatch data suggested the presence of low intensity rest tremor in more patients. This could be an indication of the higher accuracy and precision of the gyroscope signals to perceived tremor than clinical observation (Elble and McNames, 2016), or a feature of low-intensity fixed-posture tremor despite rest task examination(Louis et al., 2015). Nonetheless this consideration exceeds the purpose of this study. There is, however, a certain noise of low amplitude in signals measured by the gyroscopes that could interfere in the estimation of tremor. An exhaustive examination of the gyroscope signals showed that this basal level as ±1º/s.

Last, severity tremor rating scale was not contrasted with a rating scale of functional impairment in daily life to determine the clinically meaningful change of tremor intensity, that exceeded the scope of this study.

Further work

Further technological developments in the NetMD system is expected to simplify and automate the process to non-experts clinicians. In addition to designing algorithms to track the intensity and frequency of tremor, further developments in this area of research must include the context awareness. This will be possible by implementing automatic classification systems that are able not only to detect and quantify tremor, but to also to identify activities of the daily living (Serrano et al., 2017). Also, functional impairment in daily life and clinically meaningful change of tremor intensity should be addressed in future studies.

It is also important to note that commercial clinical uses of smart devices require compatibility with current regulatory supervision by the Food and Drug Administration in United States of America or the European Union (MEDDEV 2.1/6 July 2016; MEDDEV 2. 4/1 Rev. 9 June 2010) (Kubben et al., 2016). In European Union, smartwatch application to registry and analyses of tremor is considered a medical device class I (non-invasive). Consequently, further developments and experiments must comply with these regulations.

CONCLUSION

In this paper, we presented a full evaluation of tremor measures with smartwatches and their repeatability and correlation with clinical scores. Further studies are required to check the validity of using smartwatches as transducers in common clinical practice. However, our results show that tremor quantification by smartwatches is feasible, well-correlated with clinical scores, reliable and well-accepted by patients and, therefore, they could be another option to quantify tremor in an objective way.

Disclosures:

Dr. López-Blanco reports no disclosures.Dr. Velasco reports no disclosures.Dr. Méndez-Guerrero reports no disclosures.Dr. Romero reports no disclosures.Dr. del Castillo reports no disclosures.Dr. Serrano reports no disclosures.Dr. Benito-León reports no disclosures.Dr. Bermejo-Pareja reports no disclosures. Dr. Rocon reports no disclosures.

Funding: This work was supported by the Ministerio de Economía y Competitividad, Gobierno de España [RTC-2015-3967-1, “NetMD - Plataforma para el seguimiento de Trastornos del Movimiento”]

ACKNOWLEDGMENTS: This work was supported by the Ministerio de Economía y Competitividad, Gobierno de España [RTC-2015-3967-1, “NetMD - Plataforma para el seguimiento de Trastornos del Movimiento”]

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Figure Captions

Figure1. NetMD system

A. Subject with the set of one Smartwatch3 (SW3) Sony©, connected by Bluetooth with an Android Smartphone ASUS©, using a research version NetMD Android app of Experis IT. Ecologic fixation solution: a belt-pouch (Kalenji ®).

B. Reference system of the SW3: the three axis of space (red arrows), accelerometer axis (straight blue arrows) and the direction of the gyroscopes (curved blue arrows). Block diagram of the NetMD Android app. The app in the smartphone commands the beginning and ending of the recording. It also requests the registered data from the smartwatch and creates a TXT file to be post-processed.

Figure 2. Tremor intensity measures. Correlation of Smartwatch Mean Intensity of tremor with clinical rating. Rest tremor (RT), postural tremor (PT), kinetic tremor in finger-to-nose maneuver (KT_n), kinetic tremor in pouring water from a glass (KT_g). In (y) axes are shown log10 (mean of intensity of tremor). (x) axes clinical ratings by FTM-TRS from 0-4 points**.**

Figure 3. Bland-Altman plots to test repeatability in tremor measuring by smartwatch. Bland-Altman plots of measures obtained in Rest tremor (RT), postural tremor (PT), kinetic tremor in finger-to-nose maneuver (KT_n), kinetic tremor in pouring water from a glass (KT_g) tasks are shown. Confident intervals at 95% are given: garnet dashed lines for mean of difference or “bias” (red dashed lines); dark blue dashed lines for upper and lower limits of measurement differences (blue dashed lines). Outliers with > 3 standard deviation and clinical change between ratings in FTM-TRS ≥1 were removed from the plot.

Fig 1

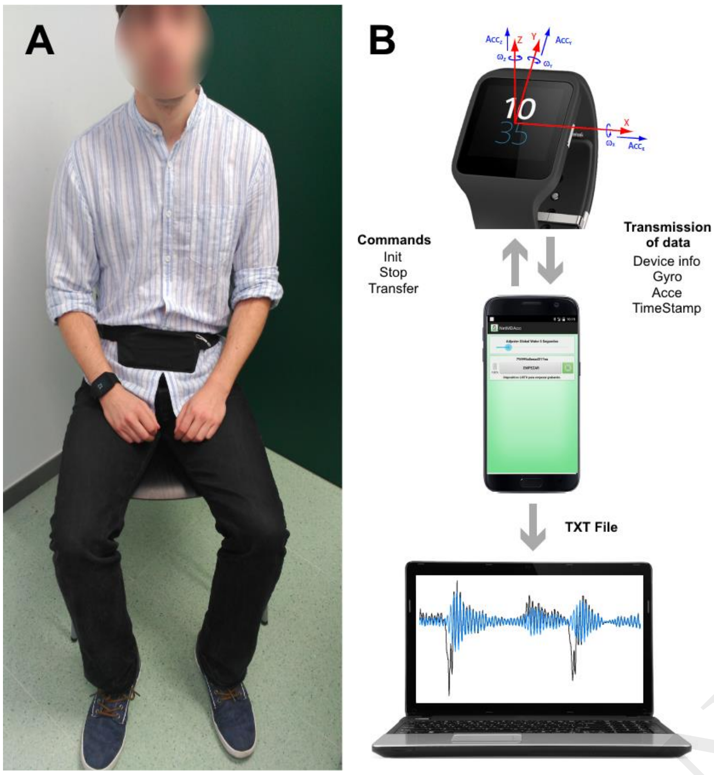
A
B
Commands
Init
Stop
Transfer
Transmission of data
Device info
Gyro
Acce
TimeStamp
TXT File

Fig 2

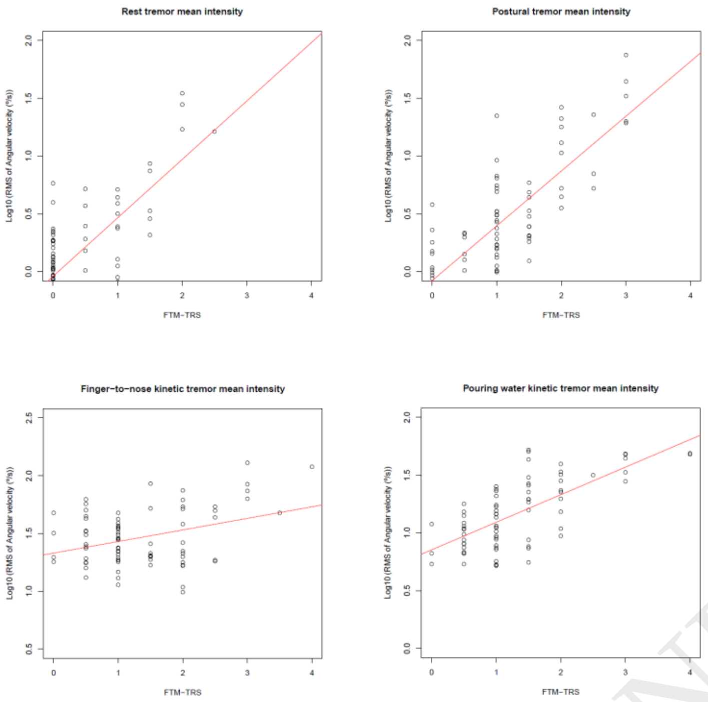
Rest tremor mean intensity
Postural tremor mean intensity
Finger-to-nose kinetic tremor mean intensity
Pouring water kinetic tremor mean intensity
Log10 (RMS of Angular velocity (°/s))
FTM-TRS

Fig 3

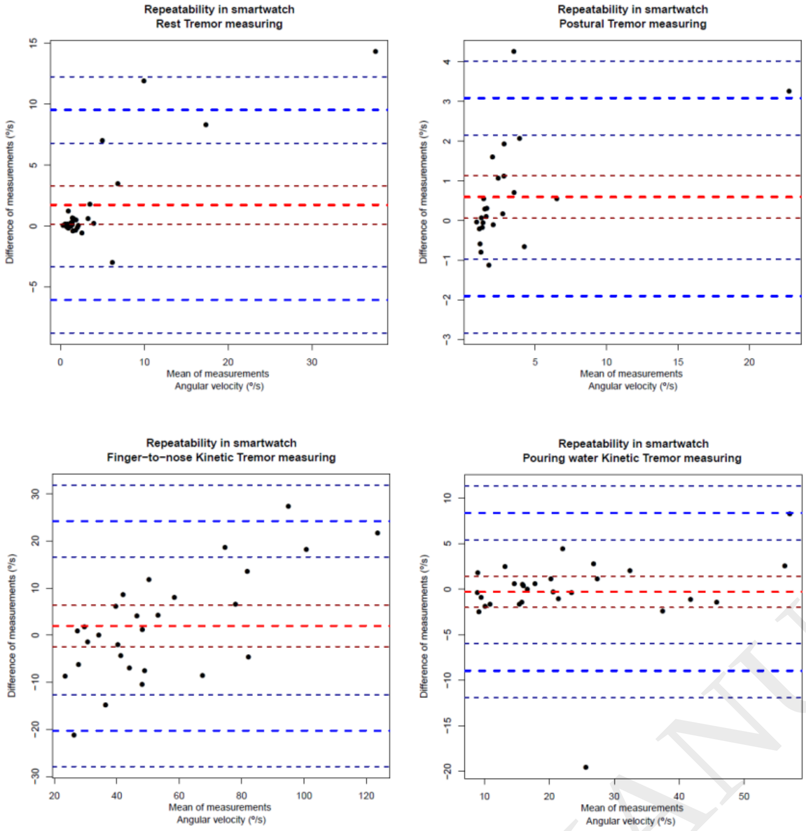
Repeatability in smartwatch
Rest Tremor measuring
Repeatability in smartwatch
Postural Tremor measuring
Repeatability in smartwatch
Finger–to–nose Kinetic Tremor measuring
Repeatability in smartwatch
Pouring water Kinetic Tremor measuring
Difference of measurements (°/s)
Mean of measurements
Angular velocity (°/s)

Table 1: Demographic and clinical of essential tremor patients (N = 34).

Gender	
Female	14 (41.1 %)
Male	20 (59.9 %)
Age at enrollment (in years)	64 ± 14.4
Disease duration	12.5 ± 10.4
Global FTM-TRS score at recruitment	27.9 ±12.6
FTM-TRS-A	9.1 ± 4.9
FTM-TRS-B	13.5 ± 6.4
FTM-TRS-C	5.3 ± 2.6
FTM-TRS score by abbreviated protocol*	
Rest tremor	0.4 ± 0.6
Postural tremor	1.1 ± 0.8
Finger-to-nose kinetic tremor	1.3 ± 0.9
Pouring water from glass kinetic tremor	1.3 ± 0.8

Mean ± standard deviation are reported. *Clinical Score relative to limb wears the smartwatch.

Parameters Tasks	Kappa	Rho	ICC	MDC=CR [º/s]	SEM [º/s]	Bias [º/s]
Rest Tremor	0.677 (p<0.001)	0.590 (p<0.001)	0.85	+/-9.94	3.58	1.74
Postural Tremor	0.661 (p<0.001)	0.738 (p<0.001)	0.95	+/-2.94	1.06	0.59
Finger-to-nose kinetic tremor	0.601 (p<0.001)	0.189 (p=0.094)	0.91	+/-24.62	8.89	1.96
Pouring water Kinetic tremor	0.557 (p<0.001)	0.652 (p<0.001)	0.95	+/-8.81	3.18	0.28

Table2. Summary of clinical correlation and test-retest reliability parameters of smartwatch tremor measures.

Kappa: Concordance between clinical raters (Linear weighted Kappa), Rho: Smartwatch measuring clinical correlation with Fahn-Tolosa-Marin Tremor Rating Scale (FTM-TRS), ICC: Intraclass Correlation Coefficient, MDC=CR: MDC (Minimum detectable change) or CR (Coefficient of repeatability) was calculated. CR = 2.77 * SEM(Vaz et al., 2013), SEM: Standard error of measurement was calculated. SEM = Standard Deviation from the 1st test x (square root of (1-ICC)), Bias: Mean of differences.

Table3. Acceptance questionnaire (adapted from Giuffrida et al(Giuffrida et al.,

Questionnaire of acceptance of the devices	YES	NO
1. Was the device comfortable to wear?	100 %	0 %
2. Did the device feel heavy?	2.9 %	97.1 %
3. Would you wear it in daily-life?	97.1 %	2.9 %
4. Did the device make difficult your arm movements?	0%	100 %
5. Did the device make difficult your trunk movements (getting up, sitting down, or walking?	0%	100 %
6. Would you find difficult the device placement at home?	12 %	88 %
7. Would you feel comfortable wearing the device in public?	83 %	17 %
8. In general, are you satisfied with the device?	100 %	0 %
9. How would you quantify your satisfaction with the device in general? From 0 (The most dissatisfaction) to 10 (the best satisfaction)	Mean = 8.7	

2009)).